

ϕ TD-Gaussian

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.06277	0.1586	0.1126	0.08364
Cramér $\ell_2^2 \downarrow$	0.0007266	0.002769	0.00282	0.001566
CF - $\ell_2^2/\omega^2 \downarrow$	0.0006073	0.00192	0.002534	0.001277

ϕ TD-Cauchy

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.1158	0.03824	0.07596	0.1349
Cramér $\ell_2^2 \downarrow$	0.003653	0.0001884	0.001761	0.002655
CF - $\ell_2^2/\omega^2 \downarrow$	0.002261	0.0001634	0.0009259	0.001525

ϕ TD-Gamma

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.714	2.458	1.944	0.07336
Cramér $\ell_2^2 \downarrow$	0.2966	0.7957	0.736	0.0002777
CF - $\ell_2^2/\omega^2 \downarrow$	0.2608	0.7209	0.6943	0.000106

ϕ TD-Categorical

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.279	0.281	0.2851	0.3112
Cramér $\ell_2^2 \downarrow$	0.02734	0.01463	0.03455	0.02856
CF - $\ell_2^2/\omega^2 \downarrow$	0.02036	0.01155	0.02845	0.02271

ϕ TD-Quantile

	Gaussian (baseline)	Cauchy mixture (heavy-tailed)	MoG (sharp / bimodal)	Gamma (skewed positive)
$W_1 \downarrow$	0.08577	0.1827	0.1059	0.1113
Cramér $\ell_2^2 \downarrow$	0.001278	0.001748	0.002708	0.001207
CF - $\ell_2^2/\omega^2 \downarrow$	0.0007907	0.002661	0.002182	0.0006976